

- b) No student at your school has ever been a participant on a television quiz show: $\forall x \forall y \neg Q(x, y)$
- c) There is a student at your school who has been a contestant on Jeopardy! and on Wheel of Fortune: $\exists x (Q(x, \text{Jeopardy}) \wedge Q(x, \text{Wheel of Fortune}))$
- d) Every television quiz show has had a student from your school as a contestant: $\forall y \exists x Q(x, y)$
- e) At least two students from your school have been contestants on Jeopardy!: $\exists x \exists y ((x \neq y) \wedge Q(x, \text{Jeopardy}) \wedge Q(y, \text{Jeopardy}))$
- 9) Let $L(x, y)$ be the statement "x loves y", where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements:
- a) Everybody loves Jerry: $\forall x L(x, \text{Jerry})$
- b) Everybody loves somebody: $\forall x \exists y L(x, y)$
- c) There is somebody whom everybody loves: $\exists y \forall x L(x, y)$
- d) Nobody loves everybody: $\forall x \exists y \neg L(x, y)$
- e) There is somebody whom Lydia does not love: $\exists y \neg L(\text{Lydia}, y)$
- f) There is somebody whom no one loves: $\exists y \forall x \neg L(x, y)$
- g) There is exactly one person whom everybody loves: $\exists y \exists z \forall x (L(x, y) \wedge L(x, z) \Rightarrow (y = z))$
- h) There are exactly two people whom Lynn loves: $\exists x \exists y \exists z ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \wedge L(\text{Lynn}, x) \wedge L(\text{Lynn}, y) \wedge \neg L(\text{Lynn}, z))$
- i) Everyone loves himself or herself: $\forall x L(x, x)$
- j) There is someone who loves no one besides himself or herself: $\exists x (L(x, x) \wedge \forall y ((x \neq y) \wedge \neg L(x, y)))$